

RAJIT GOEL

Dallas, TX, USA | US Citizen

+1 (469) 771-7869 ✉ ragoel123@gmail.com [in linkedin.com/in/ragoel](https://www.linkedin.com/in/ragoel) github.com/Drakatoa rgoel-portfolio.vercel.app

Education

Georgia Institute of Technology

Master of Science in Computer Science | Artificial Intelligence Specialization

August 2026 – December 2027

Atlanta, GA

- On-Campus Program

University of Texas at Dallas

Bachelor of Science in Computer Science | GPA: 3.965/4.0

August 2022 – May 2026

Richardson, TX

- CS² Honors Program
- Certificate in Applied Experience Design and Research
- Goldman Sachs Excellence in Computer Science Scholarship recipient
- Capital One Scholarship in Applied Experience Design and Research recipient

Professional Experience

Samsung Electronics America

June 2026 – Present

Software Engineer Intern, Networks, Systems Support / Tools & Automation

Plano, TX

- Architected an agentic AI system that autonomously diagnoses RAN network alarms end-to-end, cutting field-engineer diagnosis time from ~2 hours to 10–20 minutes by automating data collection, root-cause analysis, and remediation-step generation
- Designed a plan-and-execute agent (LLM planner → ReAct execution loop → verification) with bounded iteration, confidence gating, and human-in-the-loop escalation to keep autonomous diagnosis reliable and safe
- Built a Retrieval-Augmented Generation (RAG) pipeline over a LanceDB knowledge base of alarm runbooks and Method-of-Procedure (MOP) documents, running local LLM inference on Ollama (Qwen / Gemma) across on-prem NVIDIA A30 GPUs
- Developed a read-only MCP (Model Context Protocol) server in Python (FastMCP) that safely exposes 100+ existing network diagnostic tools to LLM agents via runtime tool discovery, with no write access to production network state
- Designed an in-platform AI chatbot for the Optimus (Django + React) system that helps field engineers retrieve documentation, images, and the right tools faster, reusing the alarm system's LLM gateway and RAG engine with Optimus auth, Redis, and Postgres-backed memory
- Benchmarked local LLMs for reasoning quality, tool-call validity, and latency against a 4–5 minute delivery SLA, and produced architecture proposals, Confluence documentation, and stakeholder presentations to align AI, tooling, and automation teams

Cinemark Theatres

June 2025 – August 2025

Data Science Intern

Plano, TX

- Analyzed theater demographic data (income, ethnicity, age) using Python (pandas, scikit-learn) and SQL to identify trends in Food & Beverage performance across 300+ locations
- Built K-Means clustering models to segment theaters and compared key KPIs (per cap, incidence rate, occupancy) across clusters to inform pricing and promotional strategies
- Conducted targeted release analysis to assess the impact of specific movie launches on F&B spending behavior across demographic segments
- Performed market basket analysis with SQL and Excel to uncover high-value product pairings and upsell opportunities, improving combo design recommendations
- Designed and delivered executive presentations with data visualizations in Power BI, Excel, and matplotlib, translating complex findings into actionable business insights
- Collaborated cross-functionally on "Pop & Top" initiative, leveraging guest surveys and international sales data to develop a business case for topped popcorn offerings

Thomson Reuters

May 2024 – August 2024

Software Engineer Intern

Frisco, TX

- Developed RESTful APIs in C# ASP.NET for tax form validation, improving efficiency and cutting processing time by 10–15%
- Refactored validation logic into modular C# components, reducing turnaround from weeks to near real-time
- Queried large datasets in SQL Server Management Studio, ensuring accurate integration of complex validation rules
- Converted legacy XML rules into modern XSLT, streamlining integration with current platforms
- Collaborated with a global Agile/Scrum team (Canada, India, Mexico, Russia), managing Git-based version control for reliability

Projects & Research

AI Sound Effects Generator - Auralis | *Next.js, Flask, PyTorch, PostgreSQL, Gemini API*

June 2025 – November 2025

- Created an AI sound generation platform that produces studio-quality effects from text prompts, powered by PyTorch AudioLDM latent diffusion models for real-time synthesis with CUDA acceleration and CPU fallback
- Integrated Google Gemini 2.5 Flash API for adaptive prompt refinement, enhancing the diversity, realism, and coherence of generated audio outputs
- Built a Next.js 15 frontend with Supabase authentication, Row Level Security (RLS), and Google OAuth, providing personalized user sessions and secure content management
- Architected a Flask REST API backend optimized for low-latency streaming, blob storage, and CORS-compliant delivery, ensuring smooth cross-browser performance
- Designed PostgreSQL schemas with triggers, analytics tracking, and indexed queries supporting a public sound library, like system, and real-time engagement metrics

AI-Powered Whiteboard - Ideate | *React, Next.js 16, TypeScript, PostgreSQL, NVIDIA Nemotron API*

November 2025

- Engineered an AI-powered ideation platform that transforms hand-drawn sketches and written concepts into structured product blueprints, including flowcharts, business pitches, competitive analyses, and 90-day roadmaps
- Integrated the NVIDIA Nemotron API (vision + text models) using a dual-model architecture to interpret sketches, extract features and user flows, and generate coherent product narratives
- Built an interactive whiteboard in React, Canvas API, and perfect-freehand, enabling precise drawing, shape recognition, undo/redo, and real-time manipulation of visual elements
- Developed Next.js 16 + TypeScript REST APIs for AI workflows including sketch analysis, Mermaid diagram generation with syntax validation and retry logic, automatic pitch creation, and competitor market analysis
- Designed PostgreSQL schemas with Supabase for persistent canvases, Row Level Security (RLS), and Google OAuth-based session management
- Deployed the full-stack application on Vercel using serverless functions for AI endpoints, optimizing performance and minimizing latency for sketch-to-insight generation
- Award: Earned top-5 recognition (Honorable Mention, NVIDIA Track) out of 100+ projects at HackUTD 2025 for innovative multimodal AI integration and end-to-end product ideation automation

SONARE.LIVE - Real-Time Lyric Performance Game | *JavaScript, Vite, Three.js, MediaPipe, WebGL/GLSL, Web Workers*

April 2026 – Ju

- Designed and built a browser-based, fully client-side real-time rhythm game for the Hatsune Miku "Magical Mirai 2026" Programming Contest, spanning ~85 ES modules with no UI framework or backend by design
- Engineered a dual-input gesture engine that unifies mouse drags and live webcam fingertip tracking into a single input stream, feeding a custom shape-recognition pipeline (normalized template matching, arc-length resampling, and scored matching) that auto-tunes difficulty per input mode
- Built a real-time hand-tracking pipeline using MediaPipe (Pose + Hand Landmarkers) with 1-Euro filtering, median despiking, and asymmetric hysteresis for reliable pen-up/pen-down detection and two-hand tracking without label-swapping
- Developed a 4-layer procedural character animation system driving VRM blendshapes (viseme lip-sync, idle blink/gaze, music-driven emotion, and pointer-relative gaze) with Three.js, custom GLSL water shaders, and spring-bone physics
- Moved MediaPipe inference off the main thread into a Web Worker with createImageBitmap frame transfer and a capability-gated fallback, protecting 3D render FPS on low-end hardware while degrading gracefully to a full mouse-only experience
- Authored a portable static build pipeline (PowerShell) that rewrites absolute asset paths to relative via regex and emits forward-slash zip entries, enabling deployment from any sub-path host (e.g., itch.io) and correct extraction on Linux servers

ArrestorIQ - Cross-Platform Flame Arrestor Analysis Platform | *React Native, Expo, Electron, TypeScript, Haxe, Azure DevOps*

January 20

- Built a cross-platform desktop and mobile application for Emerson (industry-sponsored capstone) that unifies flame arrestor test, CFD simulation, and bill-of-materials data for 10,000+ product configurations under a single validated platform
- Engineered a search and filtering system integrating uFuzzy fuzzy-matching across all product attributes, running ~5× faster than Emerson's internal proprietary search while supporting discrete and numeric-range filters
- Architected a unified filter pipeline that applies discrete and range filters before fuzzy matching on the narrowed candidate set, keeping the UI responsive with the full catalog loaded
- Designed and implemented the multi-product comparison feature and sortable, filterable table views used across the catalog and evaluation screens for side-by-side regression and pressure-drop analysis
- Led design of the team's project poster and architecture diagram, translating technical contributions for a non-technical audience of faculty and industry sponsors

Blockchain Messenger - E2E Encrypted Messaging on Custom PoW Blockchain | *React, Vite, Node.js, Express, Web Crypto API*

May 20

- Built a fully end-to-end encrypted, peer-to-peer messaging application on a custom proof-of-work blockchain implemented entirely from scratch, using no third-party blockchain or cryptography libraries
- Implemented client-side hybrid encryption (AES-256-GCM wrapped with RSA-OAEP) and ECDSA P-256 message signing via the browser Web Crypto API, ensuring private keys never leave the client
- Developed a Node.js + Express node tier with a REST API for transaction submission, SHA-256 proof-of-work mining, chain validation, and longest-valid-chain consensus across a decentralized peer network
- Enforced deterministic transaction and block validation rules (signature verification, identity binding, replay protection, PoW linkage) as the system's "smart contract" layer, backed by a 27-test automated suite

Eukarya - Creature Evolution Game | *Unity, C#, Blender, Figma*

January 2026 – May 2026

- Developed a creature evolution simulator spanning six evolutionary stages with both 2D top-down and full 3D environments, including a transition mechanic that lets the player breach the water's surface onto land
- Programmed the camera system, UI/UX, and audio architecture, including a persistent DontDestroyOnLoad music manager with string-keyed track lookup that loops themes seamlessly across scene transitions
- Implemented health, stamina, and overhealth mechanics using an edge-detection state machine to fire attack damage exactly once per instance independent of script execution order
- Built a counter-based PlayerPrefs unlock and level-select system and resolved cross-platform build issues (editor-vs-build hitbox bugs, HUD fog-tinting via a dedicated UI camera, aspect-ratio letterboxing)

PreFace - Redesigning the Hiring & Onboarding Experience | *Figma, React, UX Research* **August 2025 – December 2025**

- Partnered with Fisher Investments for ATCM 4341 (Applied Experience Design capstone) to reimagine recruitment by replacing traditional cover letters with interactive, role-specific courses
- Designed dual-sided platform: applicants complete skill-building modules earning verifiable certificates; employers review detailed assessment breakdowns across technical proficiency, soft skills, and values alignment
- Conducted user research, developed personas, created high-fidelity Figma prototypes, and built React frontend components for applicant dashboard, course progression, certificate sharing, and HR portal
- Enabled actionable hiring insights through structured assessments and match scoring, reducing candidate apathy and bias in traditional resume-based screening

Project Pawkour | *Unity, C#, UI Design, Figma, OpenGL* **August 2025 – December 2025**

- Built a third-person parkour game where players control a cat escaping from a secret laboratory through a low-poly environment
- Developed custom C# scripts for physics-based movement mechanics including running, jumping, dashing, and wall-running
- Implemented camera system with smooth following and adaptive positioning based on player movement and velocity
- Designed complete UI system including menus, HUD elements, and interactive components; scripted all UI functionality and interactions
- Created storyboards defining game narrative, level progression, and player experience flow from start to escape
- Integrated adaptive music system that intensifies based on player velocity, enhancing immersion during high-speed parkour sequences

Aegis - Privacy Shopping Browser Extension | *JavaScript, Chrome API, Supabase* **August 2025 – December 2025**

- Built a web extension for CS 4352 (HCI) that aggregates e-commerce reviews across the web, computes trust & review scores, and uses those scores to conditionally autofill or withhold sensitive user information
- Implemented popup UI, content scripts, and background service worker (Manifest v3) to inject safety indicators into shopping flows and surface visual trust cues (e.g., charts for risk levels)
- Integrated RESTful API with Supabase (PostgreSQL, authentication, RLS) for storing and retrieving review/usage data
- Designed HCI-focused UX flows including clear risk feedback, consent affordances, and data usage history views

Designing for Real Inclusion at UTD - HCI Research | *Qualitative Research, Figma, UX* **January 2025 – May 2025**

- Investigated nonbinary students' campus experiences after 28% reported not feeling a sense of belonging
- Conducted qualitative interviews revealing students discovered resources only through informal networks and existing policies lacked enforcement
- Proposed interventions including a centralized LGBTQ+ resource hub/map, anonymous feedback system for misgendering incidents, identity change guide for campus systems, and inclusive event feed with notifications
- Presented findings and design recommendations to UTD faculty and administration, emphasizing shift from performative to structural support

Leadership and Involvement

Chinese Student Association (CSA @ UTD) **January 2025 – Present**

Design Lead & Family Head

Richardson, TX

- Led creative direction for one of UTD's largest cultural organizations, designing in Figma t-shirts purchased by 200+ members and creating social media posts featured across CSA's Instagram
- Mentored "fam" group of first-year students, coordinating bonding activities, competitions, and collaborative initiatives across 50+ members

HackMate **January 2025 – May 2025**

Frontend Engineer

Richardson, TX

- A web platform designed to connect hackathon participants, enabling team formation, project collaboration, and event networking
- Developed frontend components using React, CSS, and Figma to deliver a consistent and engaging UI/UX
- Implemented navigation menus, group management tools, and contact forms integrated with backend APIs for dynamic content

Nebula Labs **August 2023 – January 2024**

UI Designer

Richardson, TX

- Designed and iterated UI prototypes in Figma for Jupiter, a platform connecting students with campus organizations
- Partnered with developers and peers to translate user research into intuitive flows, ensuring designs aligned with technical feasibility and product goals

STEMpump **June 2020 – September 2022**

Head of Media

Remote

- Led branding and marketing for a student-run STEM education nonprofit
- Created and taught a Python AI & TensorFlow course; produced educational videos and social media content
- Managed a media team and coordinated with developers to maintain consistent branding across all platforms

Technical Skills

Languages: C#, C/C++, Python, Java, JavaScript/TypeScript, R, SQL, HTML/CSS, MIPS Assembly, Verilog, XML, XSLT

Frameworks & Libraries: ASP.NET, React.js, Next.js, Node.js, Flask, Django, PyTorch, TensorFlow, scikit-learn, pandas, matplotlib, Three.js

AI / ML: Retrieval-Augmented Generation (RAG), LLM agents, Model Context Protocol (MCP), Ollama, LangGraph, vector databases (LanceDB), MediaPipe, prompt engineering

Developer Tools: Git, GitHub, Visual Studio Code, Android Studio, SQL Server Management Studio, Azure, Azure DevOps, Heroku, Vercel, Docker, Vite

Databases: PostgreSQL, MySQL, SQL Server, Redis, Supabase

Design Tools: Figma, Adobe Photoshop, Adobe Premiere Pro, AutoCAD

Other Skills: Data Analysis, Machine Learning, UI/UX Design, Interaction Design, Computer Networking, Operating Systems, Agile/Scrum, RESTful APIs, Power BI, Excel

Relevant Coursework

- | | | |
|--------------------------------|---------------------------|------------------------------|
| • Computer Science II | • Software Engineering | • Automata Theory |
| • Discrete Math I/II | • Database Systems | • Computer Graphics |
| • Linear Algebra | • Operating Systems | • Human-Computer Interaction |
| • Data Structures & Algorithms | • Digital Logic | • Interaction Design |
| • Computer Architecture | • Artificial Intelligence | |